

New Zealand Diploma in Cybersecurity		
HTCS6703	Network Security	Level: 6 Credits: 15

Student Name		Student ID	
Assessment Type		Weighting	70%
Due Date and Time		Total Marks	

Assignment		Weight	Marks	Submission Date	Time
Assessment One: Research on Adversarial Network Attacks		30%			
Assessment Two: Threat Hunting and Network Breach Simulation	Part A: Build & Configure	5%			
	Part B: Report (2500 words)	20%			
	Part C: Demonstration & Presentation	15%			

Student declaration I confirm that: • This is an original assessment and is entirely my own work. • The work I am submitting for this assessment is free of plagiarism. I have read and understood the Academic Integrity Procedure here. I have also read and understood the Student Disciplinary Statute here. • Students may use AI tools (e.g., ChatGPT, Grammarly, or similar) to support study and research efforts, such as clarifying concepts, exploring ideas, or improving understanding of subject matter. However, AI tools must not be used to generate or write any part of the final assignment deliverable. All submitted work must represent the student's own original thinking, analysis, and expression. Breaches of this requirement may be considered academic misconduct. • Where I have used ideas, tables, diagrams etc of other writers, I have acknowledged the source in every case.

<i>Students Signature:</i>	<i>Date:</i>
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Assessment mapping

Graduate Profile Outcomes	Learning Outcomes	A1	A2 - Part A	A2 - Part B	A2 - Part C
Analyse an IT environment's technology stack from a security perspective and identify issues that could impact organisational performance and business risks.	Assess different cybersecurity attacks on network services.	✓		✓	✓
	Analyse security mechanisms at different network layers to meet an organisation's security requirements .		✓	✓	✓
Apply communication, information design, teamwork, personal, and interpersonal skills, to enhance working effectiveness, efficiency, and quality outcomes in an organisational environment.	Apply interpersonal skills to manage network security efficiently.				✓

Assessment One: Research on Adversarial Network Attacks

Overview

This assignment requires students to independently research a cybercriminal or Advanced Persistent Threat (APT) group. Based on their findings, they will provide strategic recommendations to enhance the security posture of a fictional organization by addressing the threat actor's Tactics, Techniques, and Procedures (TTPs). The objective is to develop critical research skills, analytical thinking, and strategic cybersecurity recommendations.

Instructions

- You will work **individually** to complete this assessment.
- Select a known cybercriminal or APT group from public threat intelligence sources.
- Use open-source intelligence and academic literature to research the group's:
 1. Motivations and objectives
 2. Geographic and sectoral targeting
 3. Common tools and infrastructure
 4. Tactics, Techniques, and Procedures (TTPs), aligned to the MITRE ATT&CK
- Assess the potential impact of the threat actor on a fictional organisation within a sector of your choosing.
- Formulate strategic security recommendations that directly address the identified TTPs and organisational risks.

Report Guidelines

Your assessment and analysis must be **research-based** and presented in the form of a **written investigative report**. The report must demonstrate analytical rigour, strategic thinking, and a clear understanding of the chosen threat actor.

AI Usage Disclaimer

Students may use Artificial Intelligence (AI) tools to support study, idea generation, and concept clarification during their research phase. However, AI tools must not be used to generate, write, or structure any part of the submitted assignment deliverable. All submitted work must reflect the student's original analysis and expression. Use of AI beyond the permitted scope may constitute a breach of academic integrity.

Requirements

- Word count: **1,500 words ±10%**, excluding the reference list, table of contents, and other administrative sections.
- Use **formal academic and technical language** throughout.
- The report must follow a **formal structure** with clear **headings and subheadings** to guide the reader.
- Recommended report structure
 - Title Page
 - Table of Contents
 - Introduction
 - Analysis & discussion sections
 - Conclusion and strategic recommendations
 - References

Formatting and Submission

- All sources must be cited using the **APA referencing style (7th edition)**.
- Use **diagrams, tables, and models** where appropriate to support your analysis (e.g., MITRE ATT&CK mapping, Diamond Model).
- Avoid excessive narrative. Focus on **concise, evidence-based findings** and practical recommendations.
- Submit your report in **PDF or DOCX** format via **Moodle** by the due date.
- **Filename must follow this convention:**
 - StudentID_StudentName-HTCS6703_Assignment1.pdf

- Late submissions may result in a deduction of marks in line with academic policy.

Assessment criteria

Assessment Component	Criteria	Marks	Performance Indicators
Research Quality	Depth, relevance, and credibility of sources	9	Use of diverse and authoritative sources (e.g., threat intel reports, academic articles) Clear attribution and citation of data (APA style) Critical source evaluation and integration into findings
Threat Actor Analysis	TTPs, motivations, tools, and targeting	9	Clear identification of threat actor's profile Accurate mapping of TTPs to MITRE ATT&CK Insightful discussion of motivations, infrastructure, tools, and targeting patterns Application of relevant models (e.g., Diamond Model, Unified Kill Chain) where appropriate
Strategic Recommendations	Alignment to threat profile and organisational context	6	Feasible, risk-aligned recommendations Logical connection to threat actor behaviours and organisational vulnerabilities

			Consideration of mitigation, detection, and prevention controls
Structure and Presentation	Professionalism and readability	3	Clear report structure: Introduction, Analysis, Recommendations, Conclusion Effective use of tables, models, and headings Free of spelling/grammar errors; written in formal academic style
Referencing and Formatting	APA style, formatting consistency	3	Accurate in-text citations and reference list in APA 7th edition Consistent formatting, appropriate length, and naming conventions
Total		30	

Assessment Two: Threat Hunting and Network Breach Simulation

Overview

Students will collaborate in groups to design and execute a breach simulation that evaluates the effectiveness of an Endpoint Detection and Response (EDR) system. The assignment consists of two parts.

Instructions

- You will work **in groups** to complete this assessment.
- Read the statement and questions provided carefully.
- Your assessment and analysis should be research-based, and findings must be presented in the form of a written investigative report.
- The document must have a formal structure with a word count of 2500 words [+/- 10%], excluding the reference list, table of contents, or any other administrative sections.
- A recommended format for the report includes:
 1. Title Page
 2. Table of Contents
 3. Introduction
 4. Analysis & Discussion section(s)
 5. Conclusion and Recommendations
 6. References
- Submit your assignment via Moodle by the due date.
- The assignment must be formatted in APA style and include proper citations and references.
- The document should be structured with clear headings and subheadings.
- Submission must be in PDF or DOCX format.
- Ensure the document is named properly (e.g., [StudentID_HTCS6703_Assignment2.pdf](#)).
- Late submissions may result in a deduction of marks as per academic policies

Assignment breakdown

Part A – Build and Configure (5%)

- Design and develop a controlled breach simulation to analyse the security effectiveness of an EDR system.
- Use appropriate attack simulation tools and methodologies.
- Document the technical architecture, attack vectors, and detection mechanisms.
- Ensure that the environment is realistic and aligns with industry best practices.

Part B - Report (20%)

- Documentation: Compile a comprehensive Breach Attack Simulation Report detailing the simulation environment, execution process, and findings.
- Introduction: Overview of the breach simulation objectives and approach.
- Environment & Tools: Describe the simulated environment, tools used, and justification.
- Attack Execution: Detail the specific breach simulation steps performed.
- Findings & Analysis: Summarize observed security gaps and attack outcomes.
- Recommendations: Provide mitigation strategies based on findings.
- Conclusion: Final assessment and strategic insights.

Part C – Presentation & Demonstration (15%)

- Breach Simulation Demonstration: Technical walk-through of the attack execution. (~12 minutes)
- Business Stakeholder Presentation: Present findings in a non-technical format for executives. (~12 minutes)
- Clarity and Communication: Ability to present findings effectively to different audiences.

Assessment Marking Schedule

Assessment Component	Criteria	Marks	Performance Indicators
Part A – Build and Configure	Design and execution of breach simulation	5	Clear use of attack simulation tools and methodology Realistic environment aligned with best practices Documentation of architecture, attack vectors, and detection coverage
Part B – Investigative Report	Documentation Quality	3	Clarity, structure, and completeness of report APA formatting and referencing accuracy
	Introduction	2	Clear objectives and rationale for simulation Relevance to EDR evaluation goals
	Environment & Tools	3	Accurate description and justification of simulation tools and environment
	Attack Execution	3	Step-by-step detail of attack simulation Realism, correctness, and methodical execution

	Findings & Analysis	4	Identification of EDR performance, gaps, and insights Use of technical evidence and logs to support findings
	Recommendations	3	Relevant and actionable mitigation strategies- Strategic alignment to identified weaknesses
	Conclusion	2	Summary of outcomes and strategic implications
Part C – Presentation & Demonstration	Simulation Demonstration (Technical)	7	Clarity of technical walkthrough (~12 mins) Ability to explain tools, process, and results effectively
	Business Presentation (Executive)	5	Ability to communicate findings and impact in non-technical language (~12 mins) Focus on risk, impact, and decision-making relevance
	Clarity and Communication Skills	3	Adaptability of message for technical vs non-technical audiences Professionalism and clarity of delivery

Total	40	
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